navigate onto the hole's critical orbit, and there perform your controlled whirl. I discuss the critical orbit in Chapter 27. For now, suffice it to say that this is a highly *unstable* orbit. It is rather like riding a motorcycle around a very smooth volcano rim. If you balance delicately, you can stay on the rim as long as you want. When you wish to leave, a slight turn of the bike's front wheel will send you careening off the rim. When you want to leave the critical orbit, a slight rocket thrust will enable centrifugal forces to take over and send your ship careening toward the other black hole.

Once you are as close to the speed of light as you wish, you can launch yourself off a critical orbit toward your target galaxy in the distant universe (Figure 13.5).

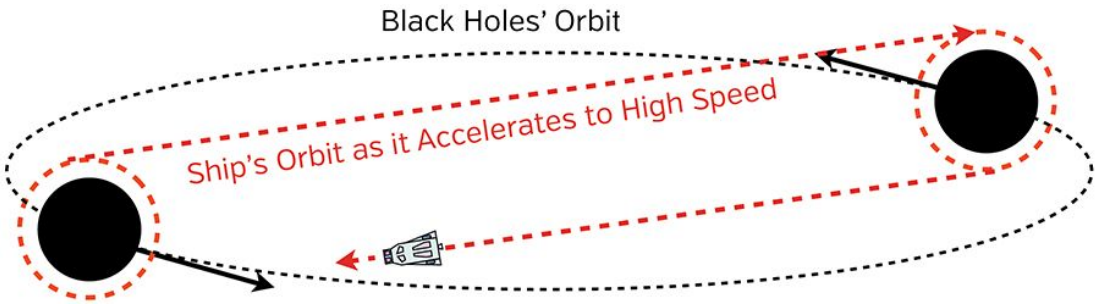


Fig. 13.4. Zoom-whirl orbit brings a spacecraft up to near light speed.

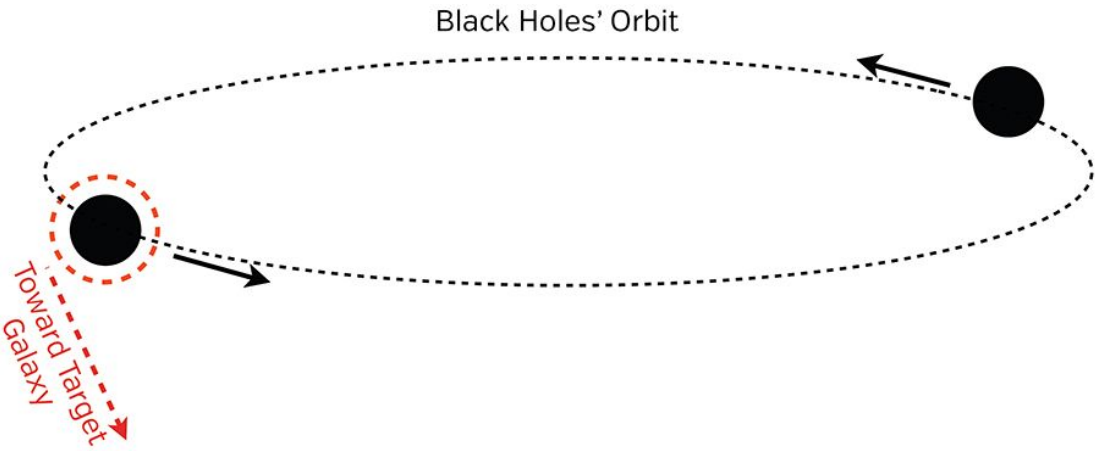


Fig. 13.5. Launching off a critical orbit toward a distant galaxy.